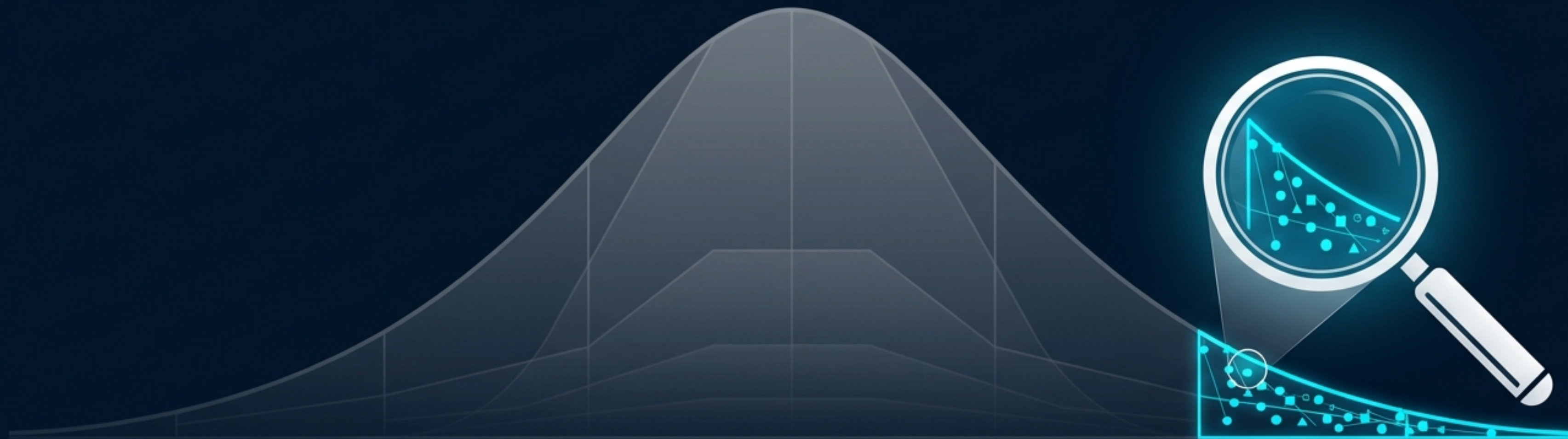


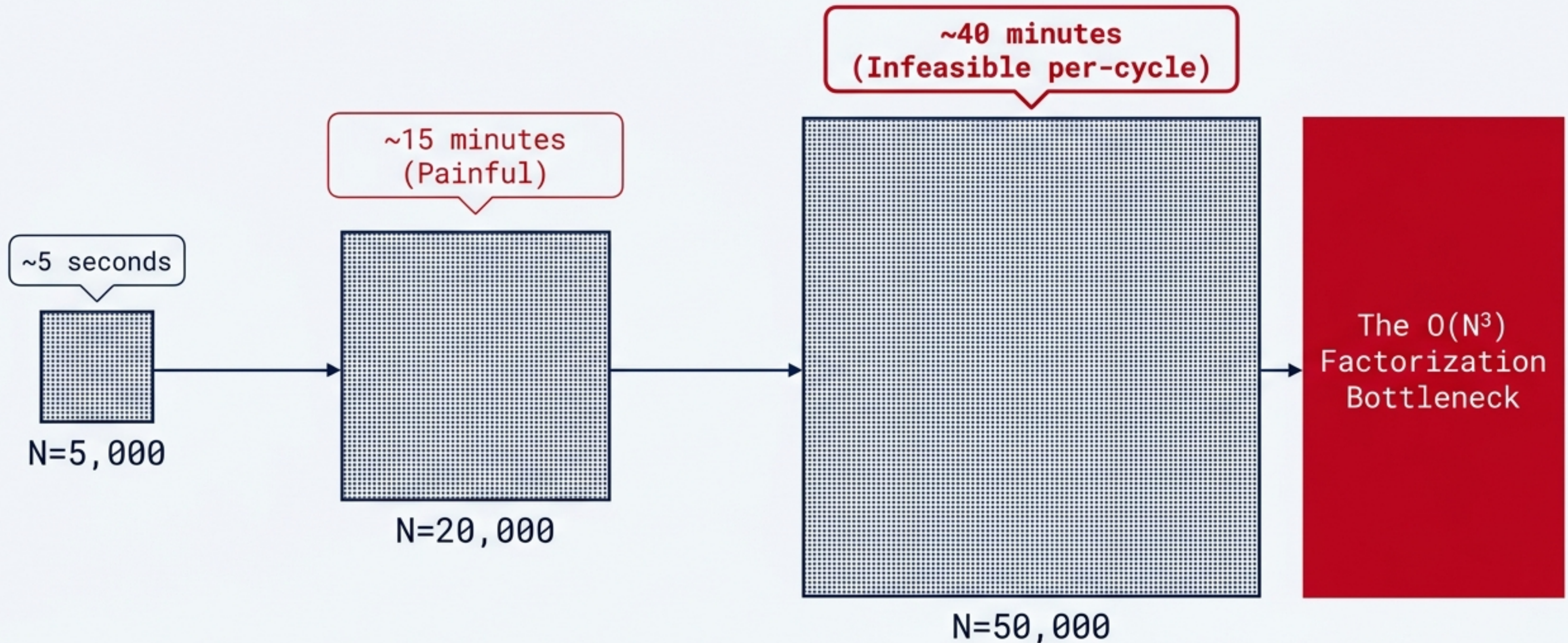
The 2% Illusion

The hidden compounding cost of the APY approximation at the elite selection tail.



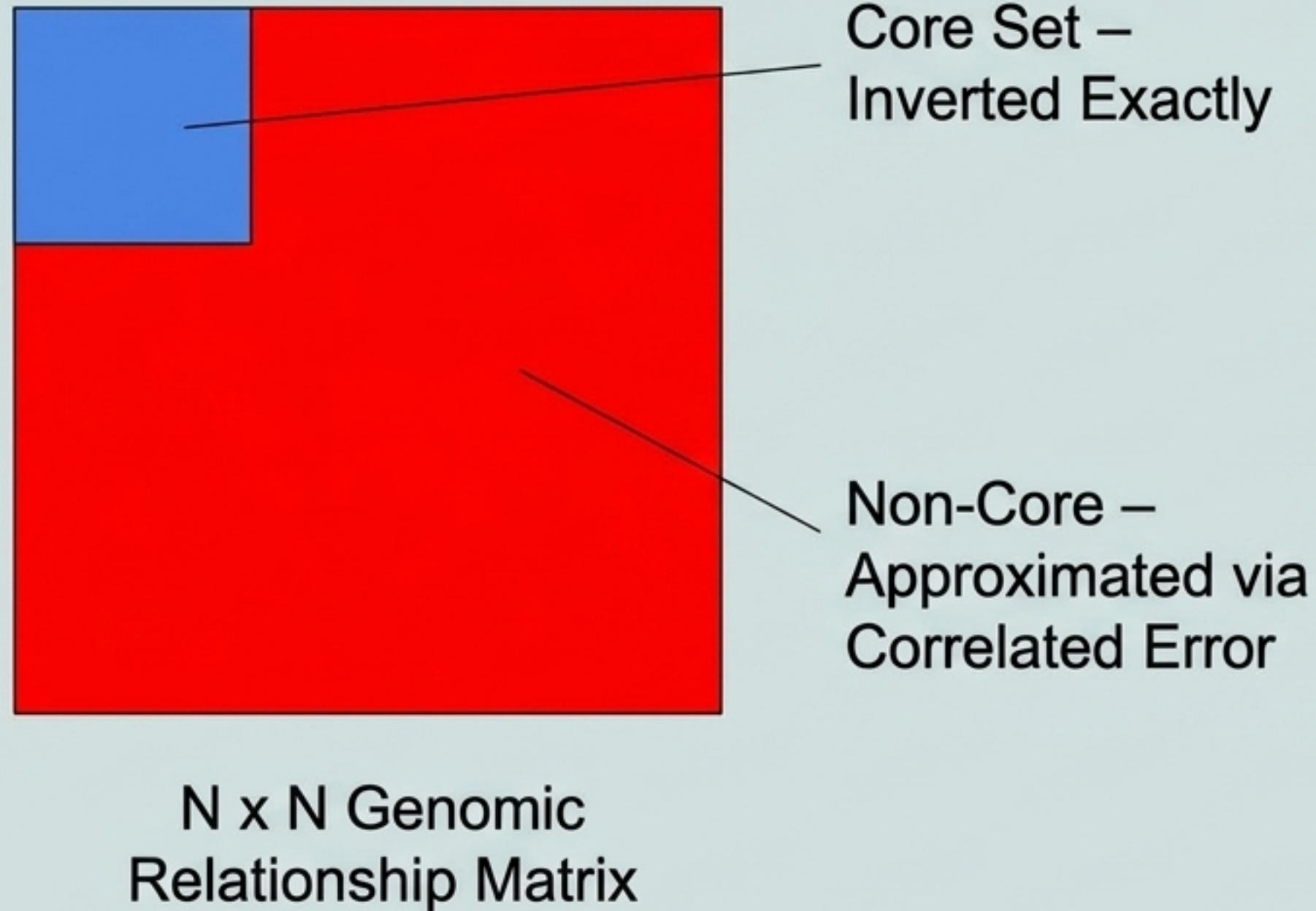
Hidden Compounding Cost: The minute variance in the extreme tail, often smoothed by APY averages, represents a significant, non-linear impact on long-term elite outcomes and risks.

The intractable physics of dense matrices forced the industry to compromise



Exact Genomic BLUP requires solving a dense, positive semi-definite linear system at every cycle. At $N > 10,000$, exact solvers break down.

The APY Workaround: Trading precision for processing speed



By partitioning animals and inverting only a small core block, the Algorithm for Proven and Young (APY) reduces an $O(N^3)$ problem to an $O(n^3)$ calculation.

APY exists out of computational desperation, not statistical superiority.

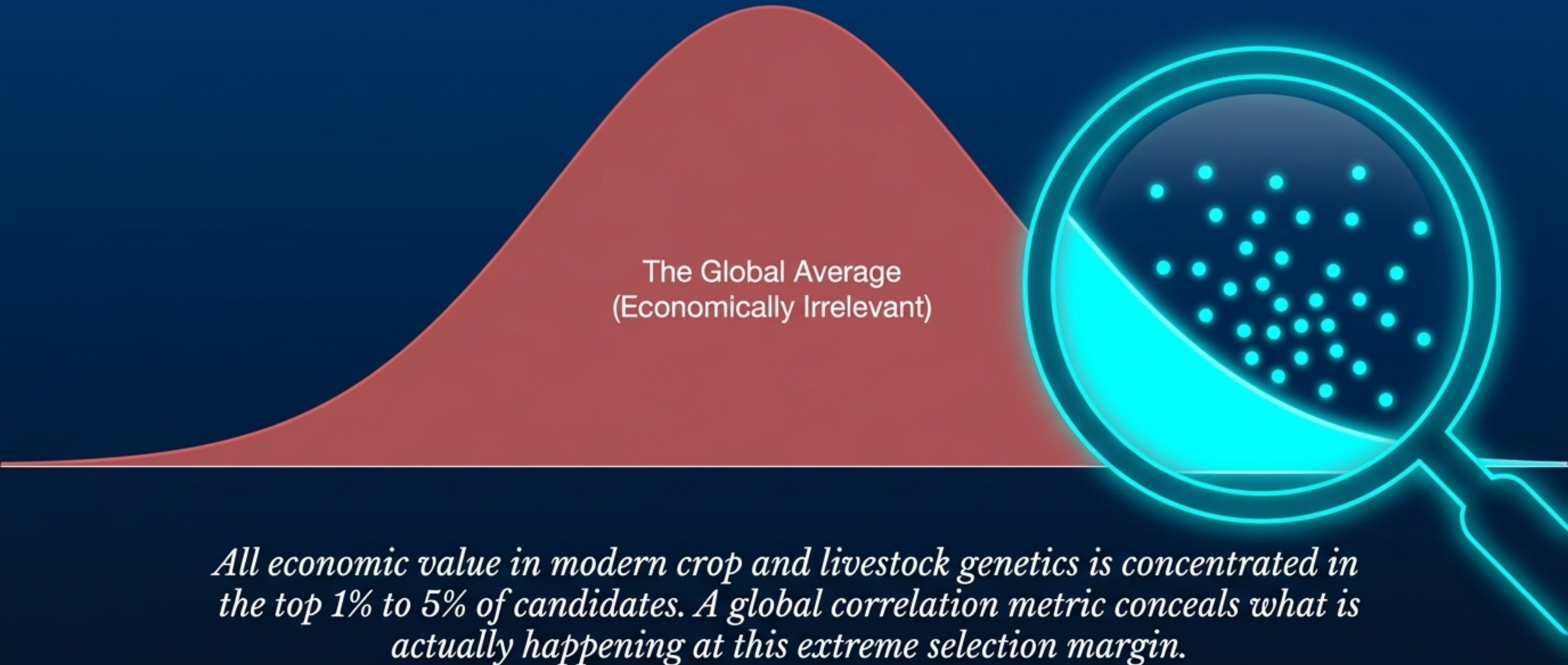


The 2% Global Accuracy Loss Claim

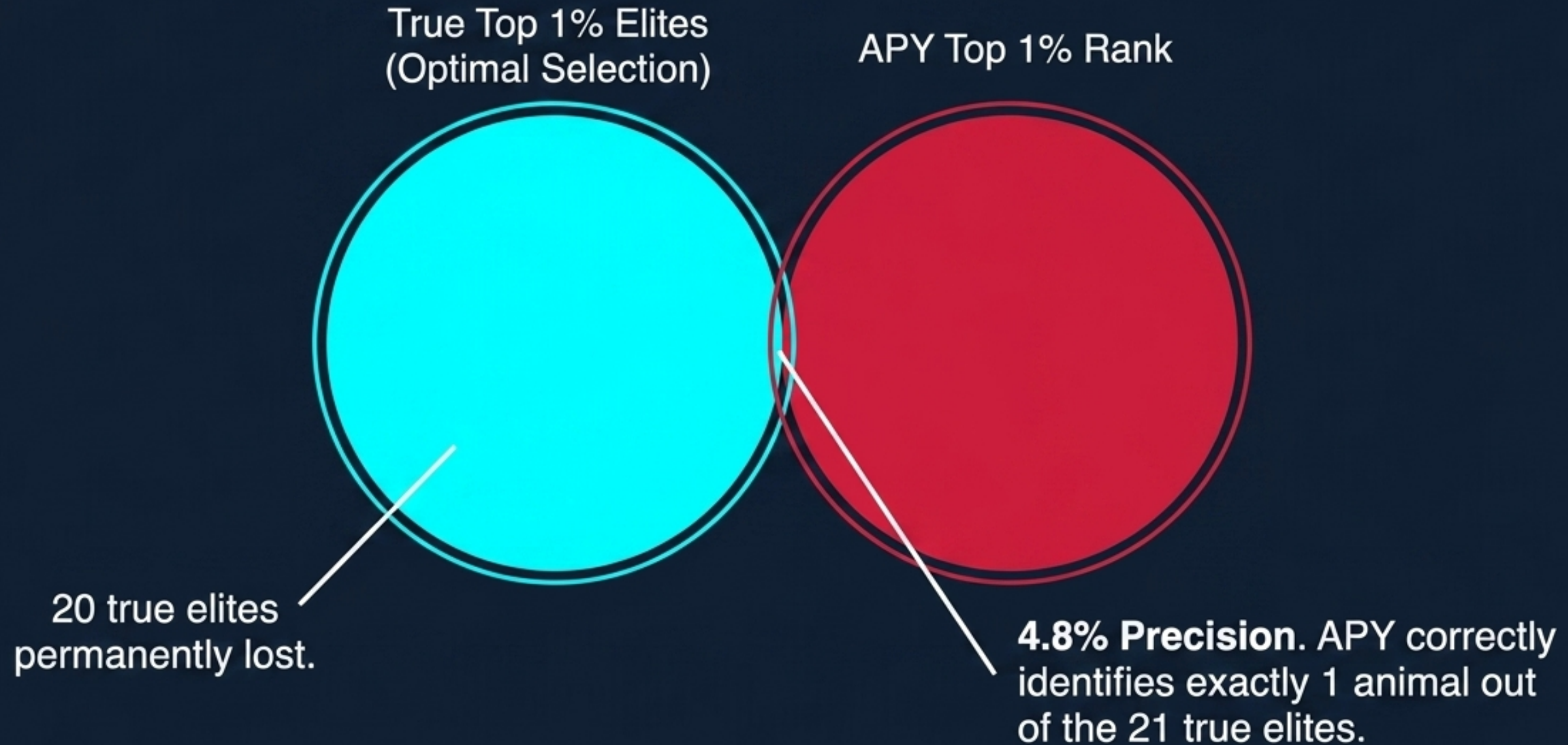
The prevailing industry headline claims APY only costs a 5-15% accuracy penalty, or a seemingly negligible ~2% drop in global Pearson correlation. Across an entire population, the approximation appears mathematically sound and economically harmless.

The comforting global average.

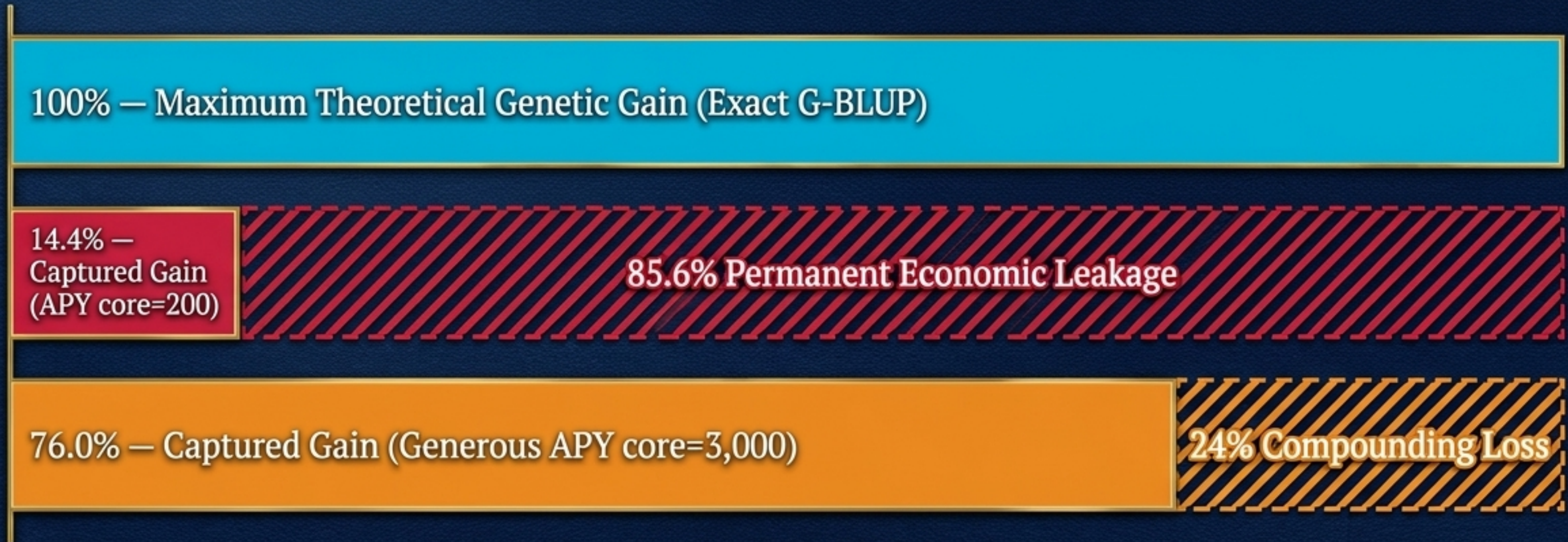
Breeding programs don't buy the average. They buy the elite tail.



Precision collapse: Losing 95% of your true elite candidates

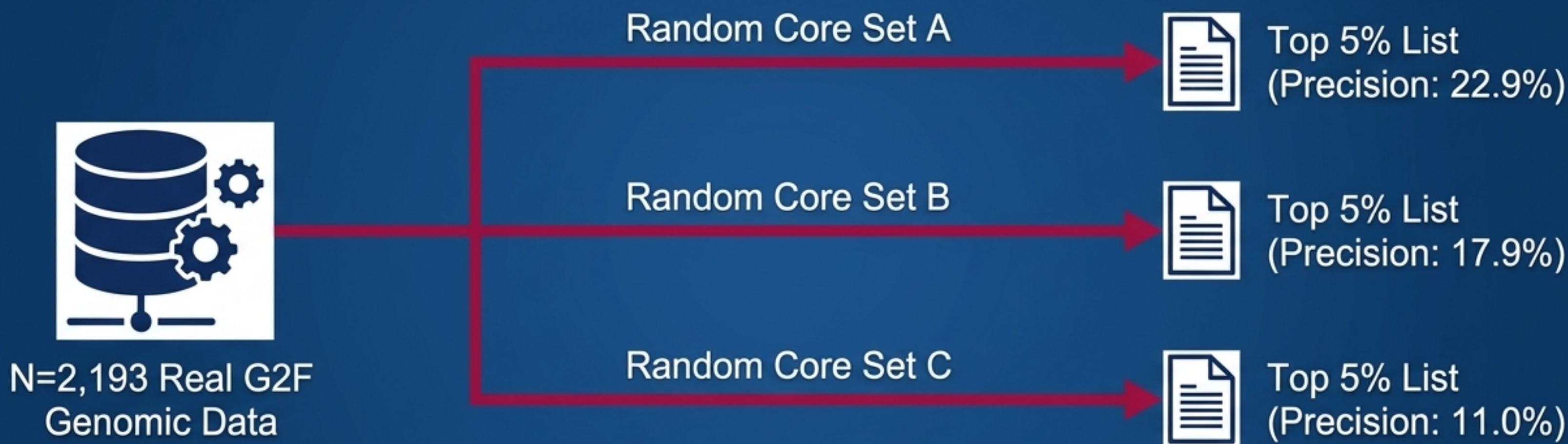


Capturing just 14 cents on the dollar of genetic potential



Selecting the top 1% via APY rank captures only 14.4% of the true breeding value sum compared to an optimal selection. This is a permanent loss that compounds across generations.

The Core-Set Lottery: Two identical programs, two completely different decisions



APY results depend entirely on the randomized core set. Two labs evaluating the exact same genomic data will **disagree** on which animals to select by a factor of 2. Precision@5% swings from 11.0% to 22.9% simply based on which random animals are assigned to the core.

The Root Cause: Correlated approximation errors across all dual coefficients

41.4x

Relative Error
at core=200

16.0x

Relative Error
even at core=1000

APY introduces **implicit, unscientific regularization** that distorts the alpha coefficients.

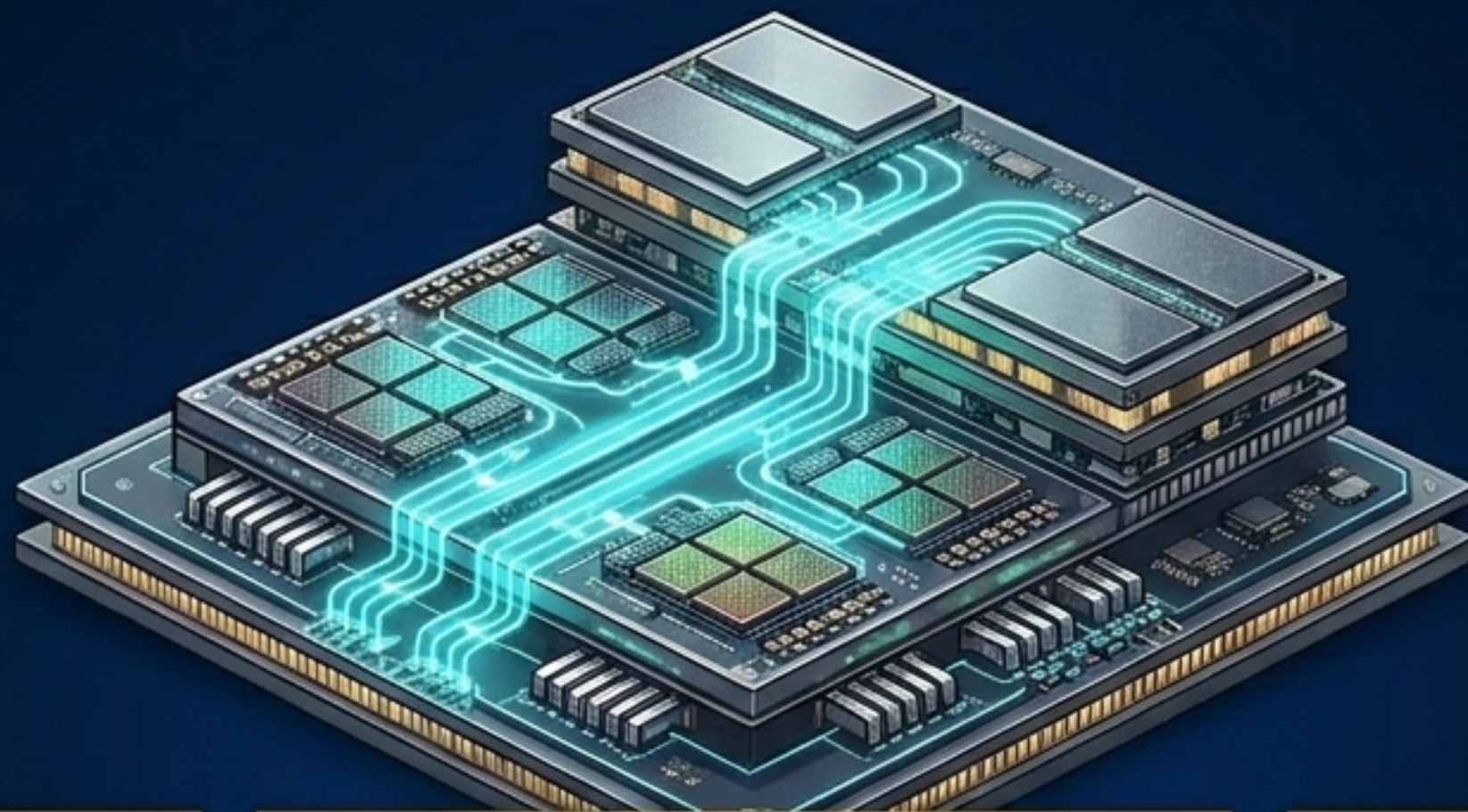
Your estimated breeding values (EBVs) are **literally 16 to 41 times wrong** before you even begin ranking them.

Exact Solution (Zero Error)

The unavoidable math of approximation vs. exactness

	APY Minimal (core=200)	APY Generous (core=1000)	Exact MPDOK
Top-1% True Elites Identified	4.8%	9.5%	100%
Max Genetic Gain Captured	14.4%	44.5%	100%
Decision Instability	$\pm 3.2\%$	$\pm 4.1\%$	0%
Coefficient Error	41x	16x	0 (Exact)
Solve Time (N=2,193)	~2.4 s	~2.4 s	86 ms

MPDOK removes the computational bottleneck. APY is no longer required.



Ideal Match

Dense Symmetric Positive Semi-Definite (SPD) matrices are MPDOK's exact target class.

TF32 LU + Float64 IR

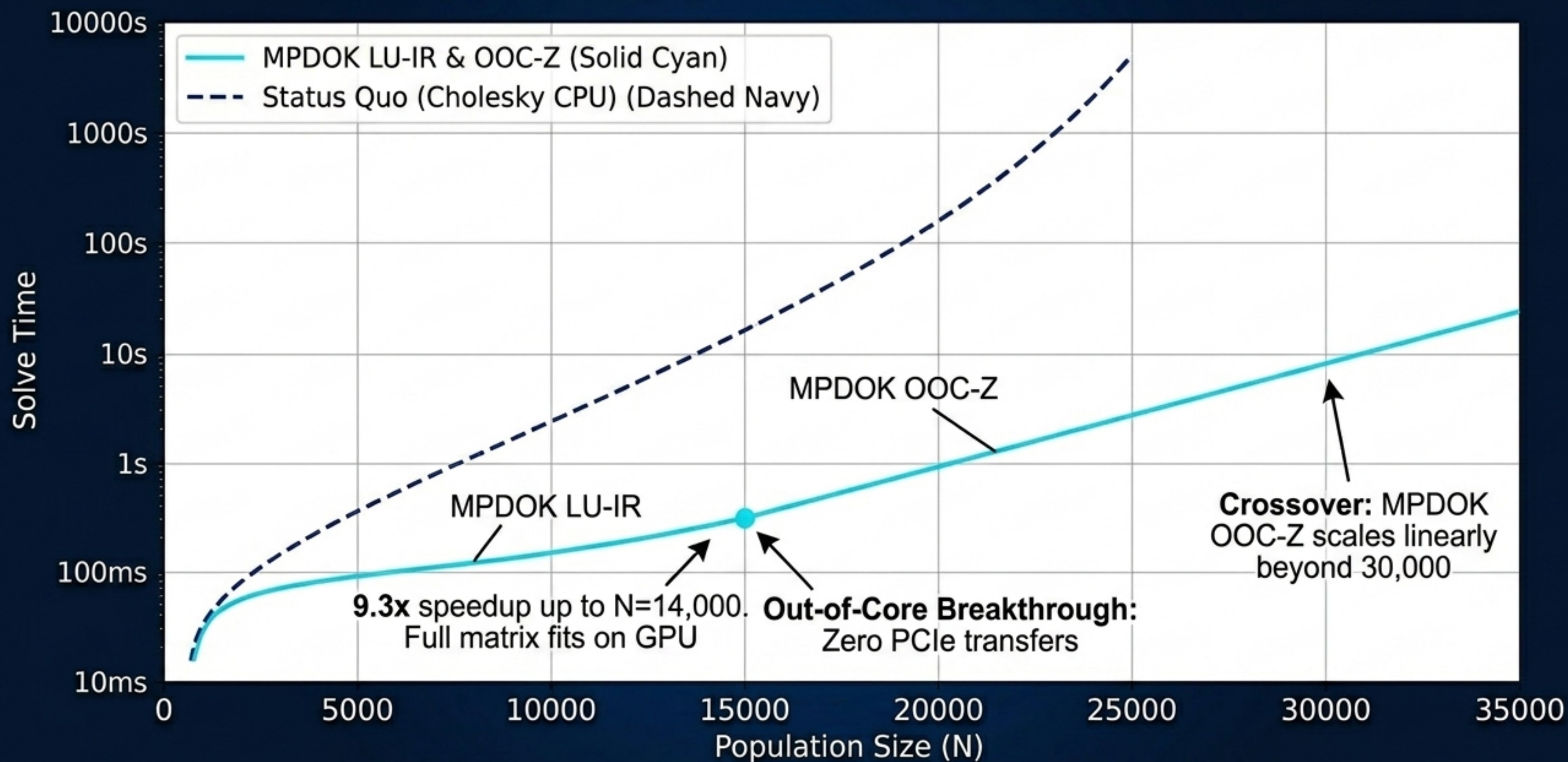
GPU TensorFloat-32 factorization paired with double-precision iterative refinement.

Zero Penalty

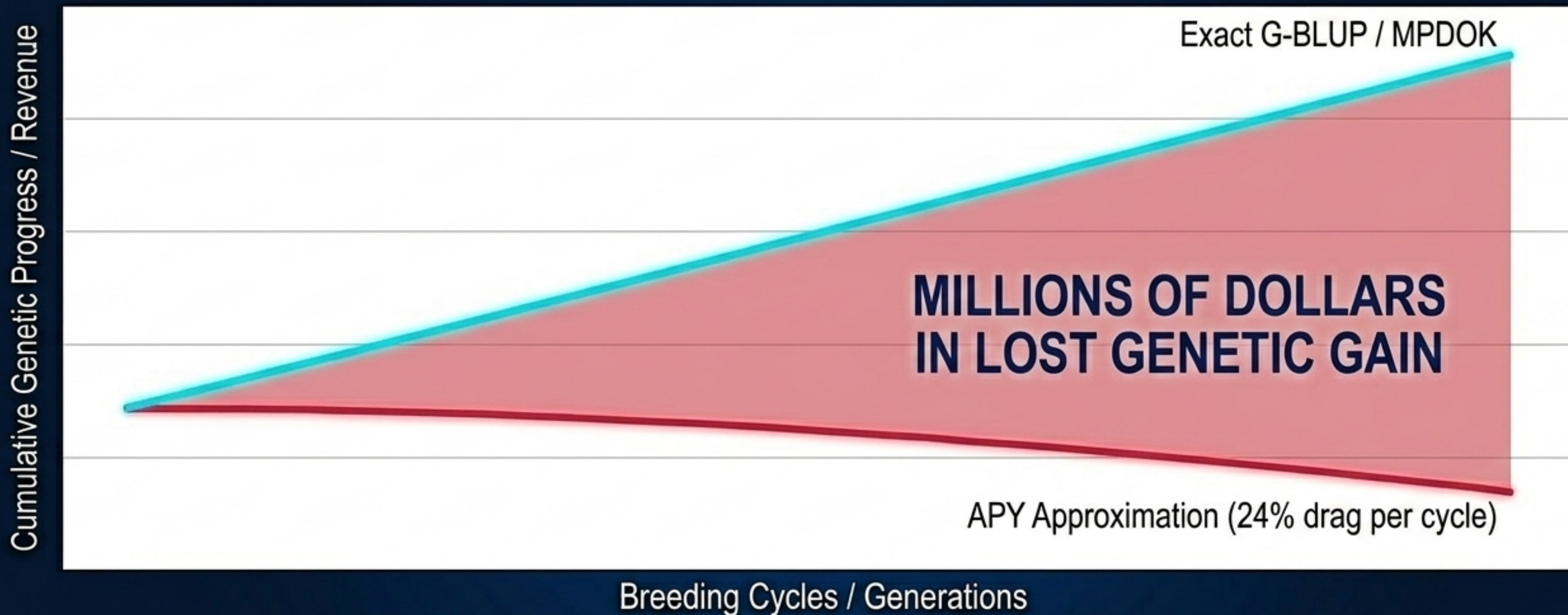
Achieves machine precision residuals ($\leq 1e-14$).

MPDOK delivers the exact G-BLUP solution at massive scale. By shifting the dense linear solve to accelerated tensor cores with iterative refinement, the $O(N^3)$ wall is shattered.

Exact solutions from the APY threshold to national population scales



The compounding cost of compromise



Genomic selection drives the future of global food security. A 5% accuracy penalty is not an engineering trade-off. It is years of lost breeding effort, missed crop yields, and permanent economic leakage. Exact solves are now tractable. **Demand them.**